

MA 600 : Research Methodology

Course Code : MA 600

Course Name : Research Methodology

L-P-T-C: 1-0-0-1

Intended for: M.Sc/Ph.D.

Prerequisites: None

Mutual Exclusion: RM of other schools

Approval : 57th BoA

Course Contents

- **Module 1: Introduction to research** (2 Lectures)
 - Defining research: Characteristics and objectives
 - Research and the scientific method
 - Various research methodologies:
 - * Descriptive vs. Analytical research
 - * Applied vs. Fundamental research o Quantitative vs. Qualitative research
 - * Conceptual vs. Empirical research
 - The research process:
 - * Formulating and defining a research problem
 - * Developing research questions
 - * Differentiating between research methods and research methodology
- **Module 2: Literature review and hypothesis development** (1 Lectures)
 - Conducting a literature review: o Reviewing concepts and theories and finding the research gaps.
 - Hypothesis development:
 - * Identifying sources and characteristics of hypotheses
 - * Understanding the role of hypotheses in research o Methods for testing hypotheses
- **Module 3: Research communication** (2 Lectures)
 - Writing and structuring research papers
 - Preparing and delivering research presentations
 - Crafting abstracts and summaries
 - Group discussion
 - Fear of rejection
- **Module 4: Research ethics** (2 Lectures)
 - Informed consent and ethical considerations
 - Maintaining confidentiality and privacy
 - Avoiding plagiarism and ensuring proper attribution

- Ethical data use and management
- Collaboration and ethical practices in joint research
- **Module 5: LaTeX for academic writing** (2 Lectures)
 - Introduction to LaTeX: Basic commands and structure
 - Writing mathematical equations and symbols
 - Organizing documents with sections, tables, and figures
 - Creating bibliographies and managing citations
 - Customizing document formats for research publications
- **Module 6: Technical writing and research documentation** (3 Lectures)
 - Writing research articles
 - Structuring and writing research projects
 - Writing theses and dissertations
 - Authoring books and writing reviews (e.g., book reviews, case reviews)
 - Understanding the criteria for high quality research
- **Module 7: Citation methods and research integrity** (2 Lectures)
 - Citation Techniques:
 - * Footnotes, endnotes, and in-text citations
 - * Compiling and formatting bibliographies
 - Ethical considerations in research and citation
 - Proper application of citation rules and guideline

Laboratory:

1. NA

Text books:

1. Wayne Goddard, Stuart Melville, **Research Methodology: An Introduction**, Juta and Co. Ltd., 2007.
2. Stefan Kottwitz, **LaTeX Beginner's Guide**, Second Edition, Packt Publishing Ltd., 2011.

References:

1. R. Barker Bausell, **Advanced Research Methodology**, Scarecrow Press, 2013.
2. Bill Taylor, Gautam Sinha, Taposh Ghoshal, **Research Methodology: A Guide for Researchers in Management and Social Sciences**, Prentice-Hall of India Private Limited, 2006.
3. Stefan Kottwitz, **LaTeX Graphics with TikZ: A practitioner's guide to drawing 2D and 3D images, diagrams, charts, and plots**, Packt Publishing Ltd. 2023.